SSZG622 - Software Project Management

Assignment 2

Topic: Schedule Management and Planning

Name: HARMALKAR RAHUL RAJAN SAYALI

Student ID: 2024tm93073

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Certification Statement

I hereby certify that the work presented in this assignment is my own and has not been copied from any other source. All external references, if used, have been acknowledged appropriately. I understand that a similarity score above 25% will result in zero marks for this submission.

Executive Summary

This report evaluates the application of **Schedule Management and Planning** in a mid-sized IT services company delivering an e-commerce platform enhancement project for a retail client. While the initial schedule was detailed using a Work Breakdown Structure (WBS) and MS Project, execution encountered setbacks due to dependency mismanagement, inadequate buffers, resource conflicts, and slow client feedback. The analysis identifies four major weaknesses: underestimation of dependencies, insufficient contingency in scheduling, poor resource allocation, and ineffective stakeholder coordination. These issues contributed to milestone delays and increased stress on delivery teams. To address these challenges, recommendations include: strengthening dependency mapping, adopting buffer management techniques, implementing resource leveling practices, and establishing a stakeholder communication plan. These align with PMBOK guidelines and best practices in Agile project delivery. By adopting structured scheduling improvements, the organization can enhance predictability, reduce delays, and improve stakeholder satisfaction in future projects.

1 - Real-life Scenario

Company Overview
The scenario is based on a mid-sized IT services firm (hereafter referred to as *TechNova Solutions*) that specializes in delivering custom enterprise applications for global clients. The company employs approximately 1,200 professionals and manages multiple parallel projects across domains such as e-commerce, banking, and healthcare. **Project Description**
The project under discussion involved the development of an **e-commerce platform enhancement** for a retail client. The scope included modernizing the client's order management system, integrating third-party payment gateways, and enabling advanced analytics dashboards. - Team Size: 35 (including developers, testers, business analysts, and a project manager)
- Duration: Initially planned for 8 months
- Methodology: Hybrid Agile (sprint-based execution with upfront scheduling using MS Project)
- Key Milestones: Requirement finalization, sprint delivery reviews, system integration, user acceptance testing (UAT), and final deployment. **Implementation of Schedule Management**
The project manager developed a **Work Breakdown Structure (WBS)** and mapped tasks into Microsoft Project to build a detailed Gantt chart. Activities were allocated across five functional teams with clearly defined dependencies. Agile practices were combined with upfront planning:
- Sprints were scheduled every 3 weeks.
- Daily stand-up meetings were conducted to monitor progress.
- Milestones were tracked bi-weekly using a project status dashboard shared with stakeholders. The schedule incorporated resource allocation charts, dependency mapping, and progress baselines. However, while the initial plan was comprehensive, challenges arose during execution.
Outcomes of Schedule Implementation
The scheduling effort allowed for clear visibility into progress and helped identify bottlenecks early. However, the project faced the following real-world outcomes:
- Delays in Milestones: Requirement finalization took 3 additional weeks due to client-side dependency.
- Integration Bottlenecks: Third-party payment gateway testing required extra time, pushing integration testing back by 2 sprints.
- Resource Conflicts: Key technical experts were shared across projects, leading to scheduling conflicts. Despite these issues, the project team was able to recover partially by fast-tracking certain tasks and reassigning developers. The final delivery was completed in 9 months—**1 month later than planned**, but within the agreed budget. **Challenges in Schedule Management**
Several challenges surfaced during the project:
1. Dependency Management – Cross-team and client dependencies were underestimated, leading to rework.
2. Inaccurate Buffering – The schedule lacked adequate contingency for integration and testing.
3. Resource Allocation Issues – Multi-project resource sharing created conflicts in availability.
4. Communication Gaps – Client feedback cycles were slower than expected, impacting milestone adherence. **Reflection**
From a personal perspective, this project reinforced the importance of building flexible schedules that accommodate uncertainty. While the project achieved its objectives, the stress on the team and slippage in delivery highlighted that effective schedule management goes beyond creating detailed plans—it requires continuous adaptation, realistic estimations, and stronger stakeholder collaboration.

2 - Analysis

Weakness 1: Underestimation of Dependencies
The project team underestimated interdependencies between modules (e.g., integration of the payment gateway and analytics dashboard). This caused rework and delays when downstream activities could not begin on time.
- Impact: Delayed integration testing and missed milestone deadlines.
- Why it was a problem: As per PMBOK, dependencies must be identified and tracked rigorously. Failure to do so creates schedule risks.

Weakness 2: Inadequate Buffering in Schedule
The initial plan had little contingency built into critical phases such as integration and testing. When client-side delays occurred, the lack of time buffers amplified schedule slippages.
- Impact: Extended project delivery by one month.
- Why it was a problem: PMBOK recommends schedule reserves to manage uncertainties. Ignoring this reduced flexibility.

Weakness 3: Resource Allocation Conflicts
Key technical resources were assigned across multiple projects simultaneously, leading to availability gaps and priority clashes.
- Impact: Some tasks remained idle while waiting for expert inputs.
- Why it was a problem: Resource leveling and allocation analysis, if applied, could have balanced workload and reduced delays.

Weakness 4: Ineffective Stakeholder Coordination
Client feedback cycles were slower than expected, but no escalation or structured communication plan was in place to mitigate delays.
- Impact: Requirements finalization slipped by 3 weeks.
- Why it was a problem: Strong stakeholder engagement is a core principle of schedule management. Lack of timely coordination hurt milestone adherence.

3 - Recommendations

1. **Strengthen Dependency Mapping**
- Use dependency matrices and critical path analysis to identify potential bottlenecks early.
- Tools such as MS Project or Jira Advanced Roadmaps can help visualize dependencies.

2. **Introduce Schedule Buffers and Risk Contingencies**
- Apply the Critical Chain Method (CCM) with project buffers to absorb uncertainties.
- Reserve at least 10–15% contingency for testing and integration phases.

3. **Implement Resource Leveling Practices**
- Use resource calendars to avoid double-booking critical experts.
- Apply load balancing across teams to ensure smooth availability.

4. **Establish a Stakeholder Communication Plan**
- Weekly status updates, escalation protocols, and sign-off checkpoints.
- Early alignment ensures faster decision-making and reduced rework.

Implementation Example: For future projects, a “Schedule Risk Review” can be held bi-weekly to track dependencies, buffer usage, and resource load. This proactive approach ensures risks are addressed before they impact milestones.

References

- Project Management Institute. (2021). *A Guide to the Project Management Body of Knowledge (PMBOK Guide) (7th ed.)*. PMI.
- Kerzner, H. (2017). *Project Management: A Systems Approach to Planning, Scheduling, and Controlling*. John Wiley & Sons.